

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element D Initial Template

Guidance before filling out the template on the next page:

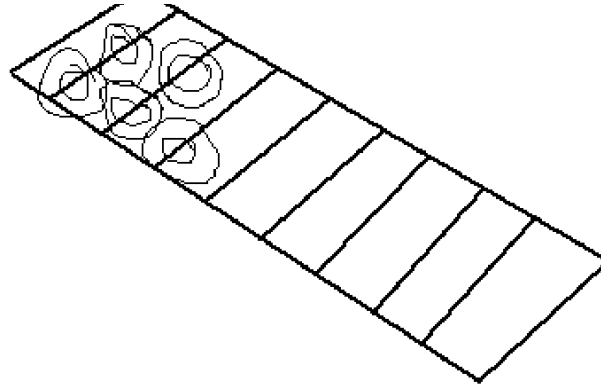
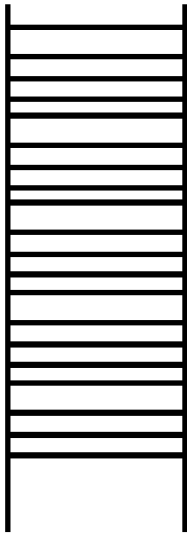
Element D: Design Concept Generation, Analysis, and Selection

Design Idea #1

Name: _____

Date: _____

Class: _____



21 x 72 in

Materials:

- Wood
- Screws & washers

Date: February 15, 2024

Description: Adding bars across the top of the shelf will allow filaments to hang across and can be moved off the top rack. Using sliding wood inside the wooden bar we can add parts that would allow for people to move a peg allowing the filament to be moved off with ease. The wooden bars can be screwed into the shelf to allow for more stability and hold more weight.

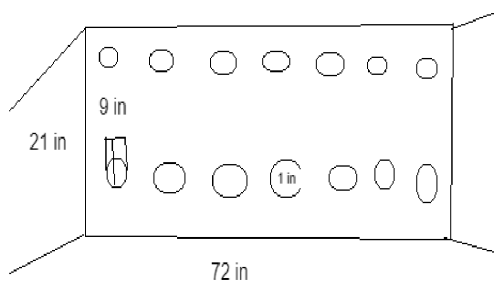


Name: _____

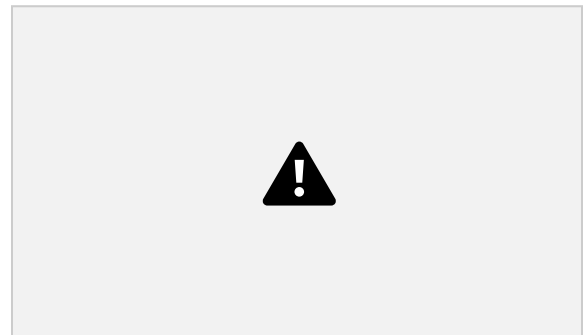
Date: _____

Class: _____

Design Idea #2



Top View



Front View

Materials:

- Wood
- Screws & washers

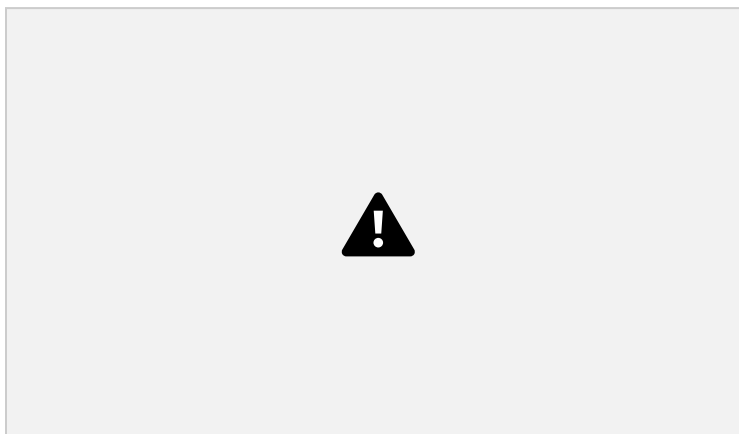
Date: February 15, 2024

Description: All the spools will be stored on the top shelf on vertical pegs organized by size, color, or whatever criterion Mr. L wants it to be organized by. The top shelf will be built from plywood and laid across the existing scaffold, which is 144 in x 21 in, so the actual plywood will need to be about an inch smaller on each side. Vertical pegs will have a diameter of 1 inch and will be spaced 9 inches apart, making a grid of 16 x 2 vertical pegs on the shelf, each peg is 16 inches tall.

Name: _____

Date: _____

Class: _____

Design Idea #3*Front View**Side View*

21 x 72 in

Materials:

- PVC Pipe
- Wood
- Screws & washers

Name: _____

Date: _____

Class: _____

Date: February 15, 2024

Description: Combining the top shelf with the vertical pegs with horizontal pegs sticking out in the front. The vertical pegs will serve as long term storage that can store the bulk of the spools. The horizontal pegs will serve as short term storage to allow students to have easy access to the spools to swap out with the existing spools. The top shelf will follow the same dimensions as idea #2, and the horizontal pegs will extrude 6 inches and be 9 inches in between. The pegs will be 1.5 inches in diameter.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Design Idea Comparison and Selection (D1, D2)

	weight	Top Shelf with Vertical Storage Pegs	Top Shelf and Horizontal T-shaped Peg	Lay Bars Along the Scaffolding	Large Coat Rack	Top Storage Shelf and Horizontal straight Pegs
Area Taken up	25	20	15	20	10	18
Weight Held	25	25	22	15	10	23
# of Stools Held	30	27	30	15	20	30
Ease of Access	20	5	12	2	20	15
score	100	77	79	52	60	86

Design Blueprint (D4)

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

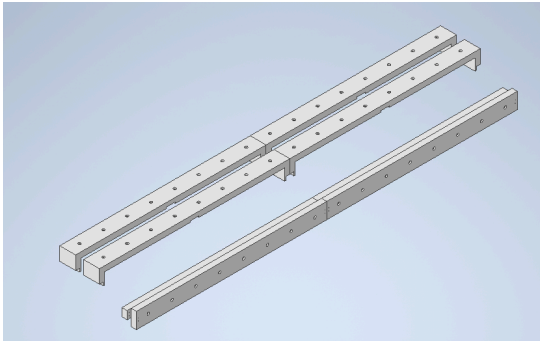


Name: _____

Date: _____

Class: _____

Design Blueprint



Sketch:

Manufacturing Methods:

1. Drill holes in the wood for the pegs
2. Place the pegs in the wood
3. Place the wood on top of the shelf
4. Drill and screw the vertical wood piece into the side of the shelf
5. Secure the screws with washers

Supply List and Costs:

Item	Cost each	# needed	Total
PVC Pipe 3/4 inch diameter	\$5.59	6	\$33.54
Wood 2 in x 4 in x 8 ft	\$3.65	6	\$21.90
Screws and washers	\$0.0	8	\$0.0
3D Printer Filament	\$0.0	96	\$0.0
Total Cost			\$55.44

Plan of Action (D3)

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include



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enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

Gantt Chart